A MONOLITHIC SOLVER FOR LINEARISED NAVIER–STOKES FREE-SURFACE DYNAMICS VIA HELMHOLTZ DECOMPOSITION

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ABSTRACT

We present a numerical implementation of the linearised Navier–Stokes (LNS) free-surface model of Galeano-Rios et al. [2017], which employs a Helmholtz decomposition of the velocity field into irrotational and vortical components. The model solves the coupled system of Laplace's equation for the velocity potential, heat equations for the vortical fields, and surface evolution equations that incorporate gravity, surface tension, and viscous effects. We implement both a sequential (operator-splitting) solver and a monolithic formulation in which all unknowns—the vortical components w^3 and w^1, the surface potential ϕ_s, the bulk potential ϕ, and the surface elevation η—are assembled into a single linear system $A\mathbf{x} = \mathbf{b}$. Because the coefficient matrix A is independent of the solution, it is factored once via sparse LU decomposition; each subsequent time step requires only $\mathcal{O}(N)$ back-substitution. We verify that the two formulations produce identical results to machine precision and demonstrate the computational advantage of the monolithic approach. We further extend the same monolithic framework to fluid–body coupling by augmenting the unknown vector with the surface pressure, vertical position, and vertical velocity of a floating rigid body, yielding a fully implicit solver in which the body motion is determined self-consistently by Newton's second law driven by gravity and the integrated fluid pressure.

Keywords Free surface dynamics · linearised Navier–Stokes · Helmholtz decomposition · monolithic solver · LU prefactorisation · fluid–body coupling · floating rigid body · finite differences

1 Introduction

The dynamics of viscous free-surface flows arise in numerous physical settings, from ocean wave propagation and droplet impact to the interaction of floating bodies with fluid baths. While the full Navier–Stokes equations govern such flows, their nonlinearity and the need to track a moving free surface make direct numerical solution computationally expensive. For problems where surface deflections are small relative to the fluid depth, a linearised approach can retain the essential physics—gravity, surface tension, and viscous dissipation—at greatly reduced cost.

Galeano-Rios et al. [2017] developed such a linearised Navier–Stokes (LNS) framework in the context of non-wetting sphere impact on a fluid bath and bouncing droplet dynamics. The central idea is a Helmholtz decomposition of the velocity field $\mathbf{u} = \nabla\phi + \mathbf{w}$, which separates the flow into an irrotational component $\nabla\phi$ satisfying Laplace's equation in the bulk and a vortical component $\mathbf{w}$ governed by the heat equation. The two components are coupled through linearised boundary conditions at the free surface, which encode the kinematic condition (the surface moves with the fluid), the dynamic condition (normal stress balance including gravity, surface tension, and viscous effects), and a vorticity evolution equation. At the bottom boundary, the non-slip condition $\mathbf{u} = 0$ couples the vortical field to the potential field.

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This framework was subsequently used by Milewski et al. [2015] to study Faraday pilot-wave dynamics, by Galeano-Rios [2016] to investigate hydrodynamic pilot-wave resonance, and more recently by Barotta et al. [2025] in the context of macroscopic Brownian motion on chaotic fluid interfaces. Agüero et al. [2022] applied the framework to the impact of a rigid sphere onto an elastic membrane, demonstrating its applicability to fluid–structure interaction problems. The classical potential-flow foundations trace back to Lamb [1932] and Batchelor [1967].

A common simplification employed in some implementations is the diagnostic relation $w^3 = (2/\mathrm{Re})\,\Delta_H \eta$ at the surface, derived by combining the kinematic and vorticity surface equations under the assumption that initial vorticity transients have decayed. While this eliminates one unknown and is valid at high Reynolds numbers, it does not capture the full transient behaviour of the surface vorticity, and it precludes correct treatment of the non-slip bottom boundary condition for the vortical field. In the present work, we solve the *full* set of equations (2.15a–e) from Galeano-Rios et al. [2017], evolving w^3 and w^1 as independent 2D fields via their respective heat equations, with proper boundary conditions at both the free surface and the solid bottom.

Our main contribution is a monolithic formulation of the full system. In the standard sequential (operator-splitting) approach, each time step involves multiple sparse solves: the heat equations for w^3 and w^1, the implicit surface update for ϕ_s, the Laplace equation for the bulk potential, and the kinematic update for η. We show that the entire system can be assembled into a single block-structured linear system

$$A\,\mathbf{x}^{n+1} = \mathbf{b}(\mathbf{x}^n),$$

where $\mathbf{x} = [w^3,\, w^1,\, \phi_s,\, \phi_{\mathrm{bulk}},\, \eta]^\top$. Because all equations are linear with constant coefficients, the matrix A does not depend on the solution. It is factored once via sparse LU decomposition at setup time, and each time step reduces to a single back-substitution at $\mathcal{O}(N)$ cost. We verify that the monolithic and sequential solvers produce identical results to machine precision, confirming the equivalence of the two formulations.

We then extend this framework to fluid–body coupling. A flat-bottomed rigid body of finite width is placed on the free surface, and its motion is determined self-consistently by Newton's second law driven by gravity and the integrated fluid pressure. Under the wetted region of the body, the standard free-surface boundary conditions are replaced by three constraints: a geometric constraint (η matches the body position), a kinematic match (the vertical fluid velocity matches the body velocity), and horizontal no-slip at the body bottom. The surface pressure p_s acts as a Lagrange multiplier enforcing the kinematic match. Augmenting the unknown vector by p_s and the two body scalars z_b, v_b, the entire coupled system remains linear with constant coefficients, so it too is amenable to a one-time LU prefactorisation. This fully implicit coupling avoids the added-mass instability that is known to afflict staggered fluid–body coupling schemes, particularly for light bodies.

The remainder of this paper is organised as follows. Section 2 presents the mathematical model: the Helmholtz decomposition, the governing equations in the bulk and at the boundaries, the non-dimensionalisation, and the extension to fluid–body coupling for a floating rigid body. Section 3 describes the numerical methods: finite difference discretisation, the sequential time-stepping algorithm, the monolithic system assembly for the pure fluid problem, and the augmented monolithic system for the coupled fluid–body problem. Section 4 presents numerical results.

2 Mathematical model

2.1 Physical setup

We consider a two-dimensional domain with horizontal coordinate $x' \in [0, L']$ and vertical coordinate z', where $z' = 0$ is at the mean free surface and $z' = -D'$ is at the solid bottom. The fluid has density ρ, kinematic viscosity ν, and surface tension coefficient σ. The free surface is located at $z' = \eta'(x', t')$, where η' is the surface elevation. Primes denote dimensional variables.

We assume that surface deflections are small relative to the depth ($|\eta'| \ll D'$), so that the free-surface boundary conditions can be linearised about $z' = 0$. Horizontal boundary conditions are periodic: $\eta'(0, t') = \eta'(L', t')$, and similarly for all other fields.

2.2 Helmholtz decomposition

Following Galeano-Rios et al. [2017], the velocity field is decomposed as

$$\mathbf{u} = \nabla\phi + \mathbf{w}, \tag{1}$$

where $\phi(x', z', t')$ is the velocity potential and $\mathbf{w}(x', z', t')$ is the vortical component. Both fields are solenoidal: $\Delta\phi = 0$ (by construction) and $\nabla \cdot \mathbf{w} = 0$ (from incompressibility). In two dimensions, $\mathbf{w} = (w^1, w^3)$ has a horizontal component w^1 and a vertical component w^3; the out-of-plane component $w^2 = 0$ by symmetry.

The appeal of this decomposition is twofold. First, the irrotational part $\nabla\phi$ satisfies Laplace's equation in the bulk—a time-independent elliptic problem that can be solved by standard sparse linear algebra once the boundary data at the free surface and the bottom are specified. Second, the vortical part $\mathbf{w}$ is governed by linear parabolic (heat) equations, which carry all of the viscous and vorticity-generating effects. For free-surface flows at moderate-to-high Reynolds numbers, the bulk flow is dominated by the potential component, while the vortical component is predominantly concentrated in thin boundary layers near the surface and the solid bottom. The decomposition thus isolates viscous effects into compact regions, leading to a system of linear partial differential equations whose structure is amenable to efficient time stepping.

2.3 Governing equations

2.3.1 Bulk equations

In the interior of the fluid domain ($z' < 0$), the potential satisfies Laplace's equation and each vortical component satisfies the heat equation:

$$\Delta\phi = 0, \tag{2}$$

$$\partial_{t'} w'^i = \nu\, \Delta w'^i, \quad i = 1, 3. \tag{3}$$

Equation (2) follows from the identity $\nabla \cdot (\nabla\phi) = \Delta\phi$ combined with the imposed solenoidal property of the potential part, and it is time-independent: the potential field adjusts instantaneously to the boundary data at the surface and the bottom. Equation (3) follows from taking the curl of the linearised Navier–Stokes equation and using the orthogonality of $\nabla\phi$ and $\mathbf{w}$; it expresses the fact that, in the bulk, the vortical field diffuses with kinematic viscosity ν and does not self-advect (a consequence of the linearisation). The coupling between ϕ and $\mathbf{w}$ is therefore entirely contained in the boundary conditions at $z' = 0$ and $z' = -D'$, presented next.

2.3.2 Free-surface boundary conditions ($z' = 0$)

Under the small-amplitude assumption $|\eta'| \ll D'$, the free-surface boundary conditions can be evaluated at the mean surface level $z' = 0$ rather than at the actual deformed surface $z' = \eta'$. This linearisation is the essential simplification that turns an otherwise nonlinear moving-boundary problem into a linear system on a fixed domain. Three physical requirements must be imposed at the free surface: a *kinematic* condition stating that the surface moves with the fluid, a *dynamic* condition balancing normal stresses at the interface, and a *vorticity* condition derived from the evolution of the tangential vortical component. In addition, a *stress-free* condition on the horizontal vortical component w^1 closes the system. All four conditions are given below in their dimensional form (cf. Galeano-Rios et al. 2017, equations 2.15c–e and 2.11a):

Kinematic condition (2.15c): The surface moves with the fluid:

$$\partial_{t'}\eta' = \partial_{z'}\phi' + w'^3. \tag{4}$$

Dynamic condition (2.15d): The normal stress balance at the free surface:

$$\partial_{t'}\phi'_s = -g\eta' + \frac{\sigma}{\rho}\,\partial_{x'x'}\eta' + 2\nu\,\partial_{x'x'}\phi'_s - 2\nu\,\partial_{z'}w'^3 - \frac{p'_s}{\rho}, \tag{5}$$

where $\phi'_s(x', t') = \phi'(x', 0, t')$ is the surface velocity potential, $\partial_{x'x'}$ is the horizontal Laplacian (which equals Δ_H in 2D), p'_s is an externally applied surface pressure (zero for an uncovered free surface), and $\kappa[\eta']$ is the mean curvature of the interface. Under the linearisation assumption, $\kappa[\eta'] \approx \partial_{x'x'}\eta'$. The four right-hand terms admit a clear physical interpretation: the first ($-g\eta'$) is the hydrostatic restoring force that drives gravity waves; the second ($\sigma\kappa/\rho$) is the capillary restoring force from surface tension; the third ($2\nu\,\partial_{x'x'}\phi'_s$) is the tangential viscous stress associated with the horizontal curvature of the potential at the surface; and the fourth ($-2\nu\,\partial_{z'}w'^3$) captures the viscous coupling between the vortical field and the surface potential via the vertical gradient of w'^3. The last term is crucial: it is the reason the surface potential and the vortical component do not decouple, even under the linearisation.

Vorticity evolution (2.15e): The surface vorticity evolves according to:

$$\partial_{t'}w'^3 = 2\nu\,\partial_{x'x'}\left(\partial_{z'}\phi' + w'^3\right). \tag{6}$$

2.3.3 Bottom boundary condition ($z' = -D'$)

The non-slip condition $\mathbf{u} = 0$ at the solid bottom requires, via the decomposition (1):

$$w'^3 = -\partial_{z'}\phi' \quad \text{at } z' = -D', \tag{7}$$

$$w'^1 = -\partial_{x'}\phi' \quad \text{at } z' = -D'. \tag{8}$$

2.3.4 Stress-free condition for w^1 at the surface

The tangential stress-free condition at the free surface (cf. Galeano-Rios et al. 2017, equation 2.11a) gives:

$$\partial_{z'} w'^1 = -\left(\partial_{x'z'}\phi' + \partial_{x'} w'^3\right) \quad \text{at } z' = 0. \tag{9}$$

Remark on equation (2.17). Combining (4) and (6) and integrating in time yields the diagnostic relation

$$w'^3 \;=\; 2\nu\, \Delta'_H \eta' \quad \text{at } z' = 0, \tag{10}$$

in the sense that the surface value of w'^3 is determined algebraically by the surface elevation once initial vorticity transients have decayed [Galeano-Rios et al., 2017]. This simplification is attractive because it eliminates one partial differential equation from the surface system; however, its validity rests on two conditions that are not always met: first, that the initial state has no residual surface vorticity, and second, that the Reynolds number is large enough that transient vorticity dynamics are irrelevant on the timescale of interest. Moreover, if the diagnostic relation is used, the vortical field loses its own degree of freedom at the surface, and the non-slip bottom boundary condition (7)–(8) can no longer be imposed independently of the potential. In the present implementation we therefore retain equation (6) as an independent evolution equation for the surface w'^3, solve the bulk heat equations (3) for w'^1 and w'^3 throughout the domain, and impose the full non-slip bottom condition separately. This design choice admits a wider range of initial conditions and correctly captures the short-time transient behaviour at the surface.

2.4 Non-dimensionalisation

We non-dimensionalise using a characteristic length L_c and velocity $V_c = L_c/T_c$:

$$x = \frac{x'}{L_c}, \quad z = \frac{z'}{L_c}, \quad t = \frac{t'}{T_c}, \quad \eta = \frac{\eta'}{L_c}, \quad \phi = \frac{\phi'}{V_c L_c}. \tag{11}$$

This introduces the dimensionless numbers

$$\mathrm{Fr} = \frac{V_c^2}{g L_c}, \qquad \mathrm{We} = \frac{\rho V_c^2 L_c}{\sigma}, \qquad \mathrm{Re} = \frac{V_c L_c}{\nu}. \tag{12}$$

The Froude number Fr measures the ratio of inertia to gravity; for $\mathrm{Fr} \gg 1$ inertia dominates and gravity waves are weakly restored, while for $\mathrm{Fr} \ll 1$ gravity strongly constrains surface motion. The Weber number We measures the ratio of inertia to capillarity; large We corresponds to a weakly capillary regime in which surface tension is unimportant at the scale L_c, whereas small We puts surface tension in the leading role. The Reynolds number Re measures the ratio of inertia to viscous diffusion; free-surface wave problems of interest typically have Re ranging from $\mathcal{O}(10^2)$ for millimetric droplets to $\mathcal{O}(10^5)$–$\mathcal{O}(10^6)$ for centimetric or larger bodies in water. These three groups are sufficient to characterise the dynamics of the pure fluid problem; an additional parameter M, introduced in Section 2.5, governs the coupled fluid–body dynamics.

The dimensionless governing equations become:

Bulk:

$$\Delta\phi = 0, \quad z \le 0, \tag{13}$$

$$\partial_t w^i = \frac{1}{\mathrm{Re}}\,\Delta w^i, \quad z \le 0, \quad i = 1, 3. \tag{14}$$

Free surface ($z = 0$):

$$\partial_t \eta = \partial_z \phi + w^3, \tag{15}$$

$$\partial_t \phi_s = -\frac{1}{\mathrm{Fr}}\,\eta + \frac{1}{\mathrm{We}}\,\partial_{xx}\eta + \frac{2}{\mathrm{Re}}\,\partial_{xx}\phi_s - \frac{2}{\mathrm{Re}}\,\partial_z w^3, \tag{16}$$

$$\partial_t w^3 = \frac{2}{\mathrm{Re}}\,\partial_{xx}\left(\partial_z\phi + w^3\right). \tag{17}$$

Bottom ($z = -D$):

$$w^3 = -\partial_z \phi, \tag{18}$$

$$w^1 = -\partial_x \phi. \tag{19}$$

Surface stress-free for w^1:

$$\partial_z w^1 = -(\partial_{xz}\phi + \partial_x w^3) \quad \text{at } z = 0. \tag{20}$$

2.5 Extension to a floating rigid body

The LNS framework extends naturally to the case of a rigid body floating on the fluid surface. We consider a flat-bottomed rectangular body of half-width a and mass per unit length m, centred at $x = x_c$, whose bottom lies at dimensional vertical position $z_b'(t')$ with vertical velocity $v_b'(t') = \mathrm{d}z_b'/\mathrm{d}t'$. At $t' = 0$ the body is in contact with the fluid surface. Unlike prescribed-motion formulations, the body motion here is determined self-consistently by Newton's second law driven by gravity and the integrated fluid pressure. This choice distinguishes the present formulation from a wavemaker-type treatment in which $z_b'(t')$ would be specified in advance: here, both the body trajectory and the fluid response are unknowns of the coupled problem, and the interaction between them is mediated by the surface pressure p_s' on the wetted region of the body.

Body equations of motion. In dimensionless form, with $M = m/(\rho L_c^2)$ denoting the dimensionless mass per unit length,

$$\frac{\mathrm{d}z_b}{\mathrm{d}t} = v_b, \tag{21}$$

$$M\,\frac{\mathrm{d}v_b}{\mathrm{d}t} = -\frac{M}{\mathrm{Fr}} + \int_{|x-x_c|<a} p_s(x,t)\,\mathrm{d}x, \tag{22}$$

where p_s is the surface pressure. The integral in (22) represents the net upward force exerted by the fluid on the wetted region of the body.

Boundary conditions under the wetted region. Let $\mathcal{B}(t) = \{x : |x - x_c| < a\}$ denote the wetted region. Under $\mathcal{B}$, the free-surface boundary conditions must be replaced because the presence of the solid body renders the interface no longer a free surface: both the location of the interface and the tangential velocity of the fluid there are now constrained by the body geometry and motion. Three independent conditions are required:

1. *Surface follows the body* (geometric constraint):

$$\eta(x,t) = z_b(t), \qquad x \in \mathcal{B}(t). \tag{23}$$

   Because the body is rigid and the surface is assumed to remain in contact with its flat bottom, the surface elevation under $\mathcal{B}$ is pinned to the body position. This is a kinematic statement about the geometry of the interface, not a dynamic one: it does not by itself determine the body motion.

2. *Kinematic match* (no-penetration at the body bottom):

$$\partial_z\phi + w^3 = v_b(t), \qquad x \in \mathcal{B}(t),\ z = 0. \tag{24}$$

   The total vertical fluid velocity at the linearised surface, $\partial_z\phi + w^3$, must equal the vertical velocity of the body to prevent interpenetration. This replaces the free-surface kinematic condition (15) under the body. Its compatibility with (23) follows from differentiating the latter in time: $\partial_t\eta = \dot{z}_b = v_b$, which coincides with (15) evaluated under the body if and only if (24) holds. Thus the two constraints are not redundant but consistent, and together they fully determine the surface motion under the body.

3. *Horizontal no-slip* at the body bottom:

$$\partial_x\phi + w^1 = 0, \qquad x \in \mathcal{B}(t),\ z = 0. \tag{25}$$

   The tangential velocity of the fluid at the body bottom must vanish because the body is rigid and moves only vertically. This replaces the surface stress-free condition (20) under the body: a stress-free interface is appropriate at an air–fluid boundary, but not at a solid wall.

Role of the surface pressure. The dynamic boundary condition (16) continues to hold at $z = 0$ everywhere, including under the body. However, its interpretation changes between the two regions. Outside $\mathcal{B}$ the surface pressure p_s vanishes (the free surface is in mechanical equilibrium with atmospheric pressure, taken as zero by convention), and (16) is an evolution equation for ϕ_s. Under $\mathcal{B}$ the surface pressure is no longer prescribed: it becomes an unknown whose role is that of a Lagrange multiplier enforcing the kinematic match (24). From this perspective, p_s under the body represents *whatever pressure distribution the fluid must exert on the bottom of the body* in order to make the fluid velocity match the body velocity at every point. This interpretation connects naturally to the standard treatment of constrained dynamics in mechanics, where Lagrange multipliers represent the internal forces required to maintain holonomic constraints.

Hydrostatic equilibrium. A useful consistency check is to consider the static limit in which the body is at rest on a quiescent fluid ($\dot{v}_b = 0$, $\partial_t \phi_s = 0$, $\mathbf{w} = 0$). In this limit, the body equation of motion (22) reduces to the force balance $M/\mathrm{Fr} = \int_{\mathcal{B}} p_s \, \mathrm{d}x$, and the dynamic boundary condition (16) reduces to $p_s = -\eta/\mathrm{Fr}$ (using $\kappa \to 0$ for a flat surface under the body). Combining these with the geometric constraint $\eta = z_b$ yields the hydrostatic equilibrium depth

$$|z_b^{\mathrm{eq}}| = \frac{M}{2a} = \frac{m}{\rho \cdot 2a}, \tag{26}$$

which is precisely the two-dimensional statement of Archimedes' principle: the body sinks until the displaced fluid mass per unit length equals the body's own mass per unit length. This result provides a quantitative target for the numerical scheme in Section 4.

Summary of the coupled problem. The full coupled fluid–body problem thus consists of the bulk fluid equations (13)–(14), the piecewise-defined surface boundary conditions summarised in Table 1, the body equations of motion (21)–(22), and the bottom non-slip conditions (18)–(19). The additional unknowns introduced by the body are the surface pressure field $p_s(x,t)$ (supported only on $\mathcal{B}$), together with the two scalar body states $z_b(t)$ and $v_b(t)$.

Table 1: Surface boundary conditions as a function of region. The dynamic condition (16) applies uniformly, but with p_s prescribed to zero outside the body and left as an unknown inside.

Region	Free surface ($x \notin \mathcal{B}$)	Wetted body ($x \in \mathcal{B}$)
Surface elevation	$\partial_t \eta = \partial_z \phi + w^3$	$\eta = z_b$
Kinematic	(evolves η)	$\partial_z \phi + w^3 = v_b$
Horizontal vortical w^1	$\partial_z w^1 = -(\partial_{xz}\phi + \partial_x w^3)$	$\partial_x \phi + w^1 = 0$
Surface pressure	$p_s = 0$	p_s unknown

3 Numerical methods

3.1 Spatial discretisation

We discretise the domain on a uniform rectangular grid with n_x points in x and n_z points in z:

$$x_j = j\,\Delta x, \quad j = 0, \ldots, n_x - 1; \qquad z_i = -D + i\,\Delta z, \quad i = 0, \ldots, n_z - 1, \tag{27}$$

with $\Delta x = L/n_x$ and $\Delta z = D/(n_z - 1)$. The index $i = 0$ corresponds to the bottom ($z = -D$) and $i = n_z - 1$ to the free surface ($z = 0$).

Standard second-order finite differences are used throughout:

- **Horizontal Laplacian** $\Delta_H = \partial_{xx}$: three-point stencil with periodic wrapping,

$$(\Delta_H u)_j = \frac{u_{j+1} - 2u_j + u_{j-1}}{(\Delta x)^2}, \quad j \pm 1 \text{ taken modulo } n_x. \tag{28}$$

- **Vertical derivative**: centred differences in the interior, one-sided at boundaries.
- **2D Laplacian**: standard five-point stencil $\Delta = \Delta_H + \partial_{zz}$.

3.2 Sequential solver

The sequential solver advances the solution from time level n to $n+1$ in four sub-steps, each involving one or more sparse linear solves.

Step 1: Evolve w^3 and w^1 (heat equations). The bulk heat equation (14) is discretised implicitly:

$$\left(I - \frac{\Delta t}{\mathrm{Re}}\,\Delta_{2\mathrm{D}}\right) w^{i,n+1} = w^{i,n}, \quad i = 1, 3, \tag{29}$$

where $\Delta_{2\mathrm{D}}$ is the discrete 2D Laplacian. After solving the bulk, the boundary conditions are overwritten:

- *Bottom* ($i = 0$): $w_j^{3,n+1} = -(\phi_{1,j}^n - \phi_{0,j}^n)/\Delta z$ and $w_j^{1,n+1} = -(D_x\,\phi_0^n)_j$, where D_x is the centred difference operator in x.

- *Surface* ($i = n_z - 1$) *for* w^3: equation (17) discretised implicitly,

$$\left(I - \frac{2\Delta t}{\mathrm{Re}}\,\Delta_H\right) w_s^{3,n+1} = w_s^{3,n} + \frac{2\Delta t}{\mathrm{Re}}\,\Delta_H\,\phi_z^n, \tag{30}$$

  where $\phi_z^n = (\phi_{n_z-1}^n - \phi_{n_z-2}^n)/\Delta z$.
- *Surface for* w^1: stress-free condition (20) discretised as $w_{n_z-1}^1 = w_{n_z-2}^1 - \Delta z\,(\phi_{xz}^n + D_x\,w_s^{3,n+1})$.

Step 2: Update surface potential ϕ_s (dynamic condition). Equation (16) is discretised implicitly in the diffusion term:

$$\left(I - \frac{2\Delta t}{\mathrm{Re}}\,\Delta_H\right)\phi_s^{n+1} = \phi_s^n - \frac{\Delta t}{\mathrm{Fr}}\,\eta^n + \frac{\Delta t}{\mathrm{We}}\,\kappa[\eta^n] - \frac{2\Delta t}{\mathrm{Re}}\,(D_z\,w^{3,n+1})_s, \tag{31}$$

where $\kappa[\eta] = \Delta_H\eta$ (linearised curvature) and $(D_z\,w^3)_s = (w_{n_z-1}^3 - w_{n_z-2}^3)/\Delta z$.

Step 3: Solve Laplace equation for bulk ϕ. With the updated surface Dirichlet condition $\phi_{n_z-1,j} = (\phi_s)_j^{n+1}$ and the bottom ghost-point condition enforcing $\partial_z\phi = -w_0^{3,n+1}$ (from (18)), we solve:

$$A_{\text{Lap}}\,\vec{\phi}^{n+1} = \vec{b}_{\text{Lap}}, \tag{32}$$

where A_{Lap} encodes the five-point Laplacian with periodic horizontal and appropriate vertical boundary conditions.

Step 4: Update surface elevation η (kinematic condition). Equation (15) is updated explicitly:

$$\eta^{n+1} = \eta^n + \Delta t\,\left(\phi_z^{n+1} + w_s^{3,n+1}\right), \tag{33}$$

where $\phi_z^{n+1} = (\phi_{n_z-1}^{n+1} - \phi_{n_z-2}^{n+1})/\Delta z$.

Each time step thus requires solving: two $n_z n_x$-sized heat systems (for w^3 and w^1), two n_x-sized tridiagonal-like surface systems (for w_s^3 and ϕ_s), and one $n_z n_x$-sized Laplace system.

3.3 Monolithic formulation

The sequential solver calls `spsolve` multiple times per step. We now show that the entire system can be assembled into a single matrix equation.

3.3.1 Unknown vector

The full state at time level $n+1$ is collected into a single vector:

$$\mathbf{x} = \begin{pmatrix} \vec{w}^3 \\ \vec{w}^1 \\ \vec{\phi}_s \\ \vec{\phi}_{\text{bulk}} \\ \vec{\eta} \end{pmatrix}, \qquad N_{\text{total}} = 3\,n_z n_x + 2\,n_x. \tag{34}$$

3.3.2 Block structure

The monolithic system $A\,\mathbf{x}^{n+1} = \mathbf{b}(\mathbf{x}^n)$ has a block lower-triangular structure:

Row block	w^3	w^1	ϕ_s	ϕ_{bulk}	η	Equation
w^3	•					Heat eq. + BCs
w^1	•	•				Heat eq. + BCs
ϕ_s	•		•			Dynamic BC (16)
ϕ_{bulk}	•		•	•		Laplace eq.
η	•			•	•	Kinematic BC (15)

The block lower-triangular structure arises because:

- The w^3 block depends only on $w^{3,n+1}$ (implicitly) and ϕ^n (explicitly, in the RHS).

- The w^1 block depends on $w^{1,n+1}$ and $w^{3,n+1}$ (through the stress-free surface BC).
- The ϕ_s block depends on ϕ_s^{n+1} and $w^{3,n+1}$ (through $\partial_z w^3$).
- The Laplace block depends on ϕ_{bulk}^{n+1}, ϕ_s^{n+1} (Dirichlet top), and $w^{3,n+1}$ (ghost point at bottom).
- The η block depends on η^{n+1}, ϕ_{bulk}^{n+1} (through ϕ_z), and $w_s^{3,n+1}$.

3.3.3 Key property: constant coefficient matrix

Since all governing equations are linear with constant coefficients, the matrix A depends only on grid parameters (n_x, n_z, Δx, Δz), the time step Δt, and the dimensionless numbers (Fr, We, Re)—not on the solution. Therefore, A is factored once at setup time via sparse LU decomposition:

$$A = LU \quad \text{(computed once)}. \tag{35}$$

At each time step, the right-hand side $\mathbf{b}(\mathbf{x}^n)$ is assembled from the current state, and the new state is obtained by back-substitution:

$$\mathbf{x}^{n+1} = U^{-1}(L^{-1}\mathbf{b}(\mathbf{x}^n)) \quad (\mathcal{O}(N) \text{ per step}). \tag{36}$$

For our grid ($n_x = 300$, $n_z = 30$), the monolithic system has $N_{\text{total}} = 27{,}600$ unknowns with approximately $134{,}400$ nonzero entries. The LU factorisation takes approximately 0.06 s and each back-substitution approximately 1.9 ms, compared to the sequential solver which requires approximately 34 ms per step.

3.4 Augmented monolithic system with body coupling

To solve the coupled fluid–body problem described in Section 2.5, we extend the monolithic approach of Section 3.3 by augmenting the unknown vector with the surface pressure field and the two scalar body states. An alternative would be to couple the fluid and body solvers in a staggered fashion, advancing the fluid with a guess for the body state and then updating the body state from the integrated fluid pressure. Such staggered schemes are known to suffer from the *added-mass instability*: for bodies whose mass is comparable to or smaller than the mass of fluid they displace, the time-lag between fluid and body updates produces unphysical oscillations that require either small time steps, subiteration, or stabilisation to suppress [Hughes et al., 1981, Donea et al., 2004]. The monolithic formulation presented here sidesteps this issue by treating the body and fluid states as components of a single implicit system, so that the fluid pressure acting on the body and the body motion affecting the fluid are resolved simultaneously at each time step. Remarkably, the resulting system retains the same property as the pure fluid system: the coefficient matrix is constant in time and can be factored once via LU decomposition, after which each time step costs only a single back-substitution.

3.4.1 Augmented unknown vector

At time level $n+1$, the augmented state vector is

$$\mathbf{x} = \begin{pmatrix} \vec{w}^3 \\ \vec{w}^1 \\ \vec{\phi} \\ \vec{\eta} \\ \vec{p}_s \\ z_b \\ v_b \end{pmatrix}, \qquad N_{\text{total}} = 3\,n_z n_x + 2\,n_x + 2, \tag{37}$$

where we have replaced the separate surface and bulk potential vectors of (34) with a single ϕ vector covering all grid points (the surface row at $i = n_z - 1$ carries the values formerly denoted ϕ_s), and introduced three new components: the surface pressure $\vec{p}_s \in \mathbb{R}^{n_x}$ (with $p_{s,j} = 0$ enforced for $j \notin \mathcal{B}$), and the scalar body states $z_b, v_b \in \mathbb{R}$.

3.4.2 Piecewise surface boundary conditions

Let $\chi_j = 1$ if node $j \in \mathcal{B}$ and zero otherwise. The surface rows of the monolithic matrix differ between free and wetted nodes according to Table 1:

η rows. For free j: the discretised kinematic condition (15),

$$\eta_j^{n+1} - \frac{\Delta t}{\Delta z}\left(\phi_{n_z-1,j}^{n+1} - \phi_{n_z-2,j}^{n+1}\right) - \Delta t\, w_{n_z-1,j}^{3,n+1} = \eta_j^n. \tag{38}$$

For body j: the geometric constraint (23),

$$\eta_j^{n+1} - z_b^{n+1} = 0. \tag{39}$$

w^1 **surface rows.** For free j: the stress-free condition (20). For body j: the horizontal no-slip condition (25),

$$w^{1,n+1}_{n_z-1,j} + \left(D_x\,\vec{\phi}^{n+1}_{n_z-1}\right)_j = 0, \tag{40}$$

which couples the surface vortical component to the surface potential via the discrete first-derivative operator D_x.

p_s **rows.** For free j: the prescribed value

$$p^{n+1}_{s,j} = 0. \tag{41}$$

For body j: the kinematic matching condition (24),

$$\frac{\phi^{n+1}_{n_z-1,j} - \phi^{n+1}_{n_z-2,j}}{\Delta z} + w^{3,n+1}_{n_z-1,j} - v^{n+1}_b = 0. \tag{42}$$

These rows serve dual purposes: pinning the pressure to zero outside the body, and enforcing the no-penetration constraint (with p_s as the Lagrange multiplier) inside.

Surface dynamic BC (uniform). The dynamic condition (16) is discretised uniformly for all surface nodes j, with the p_s term always present:

$$\left(I - \frac{2\Delta t}{\mathrm{Re}}\,\Delta_H\right)\phi^{n+1}_{n_z-1,j} + \frac{2\Delta t}{\mathrm{Re}}\,\frac{w^{3,n+1}_{n_z-1,j} - w^{3,n+1}_{n_z-2,j}}{\Delta z} + \Delta t\,p^{n+1}_{s,j} = \phi^{n}_{n_z-1,j} - \frac{\Delta t}{\mathrm{Fr}}\eta^n_j + \frac{\Delta t}{\mathrm{We}}\kappa[\eta^n]_j. \tag{43}$$

For free j, $p^{n+1}_{s,j} = 0$ is enforced by the separate p_s row above, so this reduces to (31).

3.4.3 Body dynamics rows

Implicit Euler discretisation of (21)–(22) yields two additional rows:

$$z^{n+1}_b - \Delta t\,v^{n+1}_b = z^n_b, \tag{44}$$

$$M\,v^{n+1}_b - \Delta t\,\Delta x \sum_{j\in\mathcal{B}} p^{n+1}_{s,j} = M\,v^n_b - \frac{\Delta t\,M}{\mathrm{Fr}}. \tag{45}$$

The sum in (45) is the midpoint approximation to the integrated fluid pressure over the wetted region. These two rows close the augmented system.

3.4.4 Block structure and matrix properties

The augmented matrix retains a block lower-triangular structure with respect to the main dependencies, with off-diagonal coupling blocks between ϕ and $\{w^3, p_s, \eta\}$, and between v_b and p_s (via the pressure integral). Crucially, all coefficients are determined by grid parameters, Δt, the dimensionless numbers (Fr, We, Re), the body parameters (M, a, x_c), and the time-independent body mask χ. None depend on the solution. Therefore, the augmented matrix is factored once via sparse LU decomposition at setup time, and each subsequent time step reduces to a single back-substitution. The augmented system has $N_{\mathrm{total}} = 3\,n_z n_x + 2\,n_x + 2$ unknowns, increased only by $n_x + 2$ compared with the pure fluid system. For a representative production grid of $n_x = 400$ and $n_z = 80$, the augmented system contains approximately 97,000 unknowns, a modest increase from the approximately 96,000 unknowns of the pure fluid system; the two additional body rows and the n_x pressure rows contribute only marginally to the computational cost.

Algorithmic summary. Each time step of the coupled solver proceeds as follows:

1. From the state $(w^{3,n}, w^{1,n}, \phi^n, \eta^n, p^n_s, z^n_b, v^n_b)$, assemble the right-hand-side vector $\mathbf{b}(\mathbf{x}^n)$ using explicit contributions from ϕ^n, η^n, z^n_b, and v^n_b according to the derivations in this section.
2. Compute the new state $\mathbf{x}^{n+1} = \mathrm{LU}^{-1}\mathbf{b}(\mathbf{x}^n)$ by back-substitution against the pre-factored LU decomposition of A.
3. Extract the individual fields from $\mathbf{x}^{n+1}$ using the offsets in (37), and proceed to the next step.

The LU factorisation is performed only once, at simulation setup, and reused for the entire run. The matrix assembly and factorisation together constitute a one-time cost of $\mathcal{O}(N^{3/2})$ (typical for sparse direct solvers on banded systems), while each subsequent step costs $\mathcal{O}(N)$ via the back-substitution.

4 Results

5 Discussion

6 Conclusions

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